Nonlinear Modelling and Open-Loop Simulation of a Quadrotor

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Abstract

This report develops a nonlinear six-degree-of-freedom model of a quadrotor and implements it in MATLAB/Simulink for open-loop analysis. The model uses a body-fixed frame for translational and rotational dynamics and an inertial frame for position. Individual motor commands are mapped to propeller angular speeds through first-order actuator dynamics; thrust and aerodynamic reaction torque are then modelled as quadratic functions of propeller speed. The four rotor contributions are combined to obtain collective thrust and roll, pitch, and yaw moments. The resulting twelve-state model is intended to demonstrate how collective and differential rotor commands generate vertical, lateral, longitudinal, and yaw motion. Placeholders are included for simulation results, which will be populated after the Simulink test cases are completed.

1 Introduction

Quadrotors achieve translational and rotational motion by changing the speeds of four fixed-pitch propellers. Unlike a conventional aircraft, a quadrotor does not normally generate an independent horizontal control force. Instead, it rolls or pitches so that its thrust vector acquires a horizontal component. Consequently, a useful simulation model must represent the relation between rotor commands, propeller forces, body torques, vehicle attitude, and motion in an inertial frame.

The formulation in this report follows the reference-frame, kinematic, and rigid-body development presented by Beard [**?**]. The report is restricted to modelling and open-loop simulation; sensor models, state estimation, and feedback-control design are outside its scope. The principal objectives are to:

1) define consistent inertial and body coordinate frames;

2) derive the nonlinear twelve-state equations of motion;

3) augment the rigid-body model with motor and propeller dynamics;

4) implement the equations as a nonlinear state-space system in Simulink; and

5) interpret the open-loop relation between rotor commands and vehicle motion.

2 Modelling Assumptions

The quadrotor is treated as a rigid, symmetric body whose centre of mass coincides with the origin of the body frame. The inertia matrix is diagonal, the four arms have equal length, and the four propellers are identical. The thrust axis of each propeller is parallel to the body vertical axis. Ground effect, blade flapping, propeller–propeller interaction, structural flexibility, wind, and translational aerodynamic drag are neglected in the baseline model. Gravity is uniform, and a flat-Earth inertial frame is used. These assumptions are suitable for studying the dominant open-loop motion near hover, but they limit the accuracy of the model during aggressive or high-speed flight.

3 Reference Frames and Attitude

3.1 Inertial and body-fixed frames

The inertial frame $\mathcal{F}^i = \{\hat{\boldsymbol{i}}^i, \hat{\boldsymbol{j}}^i, \hat{\boldsymbol{k}}^i\}$ is fixed to the Earth and follows a north–east–down convention. Altitude h is positive opposite to $\hat{\boldsymbol{k}}^i$. The body frame $\mathcal{F}^b = \{\hat{\boldsymbol{i}}^b, \hat{\boldsymbol{j}}^b, \hat{\boldsymbol{k}}^b\}$ is rigidly attached to the vehicle, with x_b forward, y_b right, and z_b downward. Thus, the thrust generated by the propellers acts along $-z_b$.

The inertial position and body-resolved velocity are

$$\boldsymbol{p}^i = \begin{bmatrix} p_n & p_e & -h \end{bmatrix}^T, \qquad \boldsymbol{v}^b = \begin{bmatrix} u & v & w \end{bmatrix}^T. \tag{1}$$

The body angular velocity is

$$\boldsymbol{\omega}^b = \begin{bmatrix} p & q & r \end{bmatrix}^T, \tag{2}$$

where p, q, and r are the instantaneous angular-velocity components about the current body axes. They are not generally equal to the Euler-angle derivatives.

3.2 Euler rotations and transformation matrix

The attitude is parameterized using yaw ψ, pitch θ, and roll ϕ. With the convention of Beard [**?**], the transformation from body-resolved vector components to inertial components is

$$R_b^i = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}, \tag{3}$$

where $c_\alpha = \cos\alpha$ and $s_\alpha = \sin\alpha$. Since a rotation matrix is orthogonal,

$$R_i^b = (R_b^i)^T = (R_b^i)^{-1}. \tag{4}$$

3.3 Derivative in a rotating frame

A vector differentiated in the inertial frame differs from the derivative of its components observed in the rotating body frame. The transport theorem gives

$$\left(\frac{d\boldsymbol{a}}{dt}\right)_i = \left(\frac{d\boldsymbol{a}}{dt}\right)_b + \boldsymbol{\omega}^b \times \boldsymbol{a}. \tag{5}$$

The cross-product term accounts for the rotation of the body-frame basis vectors. It is fundamental to both the translational and rotational rigid-body equations.

4 Nonlinear State-Space Model

4.1 State and input definitions

The rigid-body state is selected as

$$\boldsymbol{x} = \begin{bmatrix} p_n & p_e & h & u & v & w & \phi & \theta & \psi & p & q & r \end{bmatrix}^T. \tag{6}$$

The generalized force input is

$$\boldsymbol{u}_c = \begin{bmatrix} F & \tau_\phi & \tau_\theta & \tau_\psi \end{bmatrix}^T, \tag{7}$$

where F is the positive magnitude of total thrust and the remaining terms are body moments. The nonlinear plant is therefore

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}_c), \qquad \boldsymbol{y} = \boldsymbol{x}. \tag{8}$$

4.2 Translational kinematics

Equation (3) transforms body velocity into inertial position rate. Accounting for altitude being positive upward gives

$$\begin{bmatrix} \dot{p}_n \\ \dot{p}_e \\ \dot{h} \end{bmatrix} = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ s_\theta & -s_\phi c_\theta & -c_\phi c_\theta \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix}. \tag{9}$$

At level attitude, $\dot{p}_n = u$, $\dot{p}_e = v$, and $\dot{h} = -w$.

4.3 Euler-angle kinematics

The relation between body rates and Euler-angle derivatives is

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & s_\phi \tan\theta & c_\phi \tan\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi / c_\theta & c_\phi / c_\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \tag{10}$$

The mapping becomes singular at $\theta = \pm\pi/2$. This limitation is acceptable for the near-hover simulations considered here; a quaternion attitude representation would be preferable for large rotations.

4.4 Translational dynamics

Newton's law in the inertial frame is $m(d\boldsymbol{v}/dt)_i = \boldsymbol{f}$. Applying (5) and resolving all terms in the body frame yields

$$\dot{\boldsymbol{v}}^b = \frac{1}{m}\boldsymbol{f}^b - \boldsymbol{\omega}^b \times \boldsymbol{v}^b. \tag{11}$$

Gravity expressed in the body frame is

$$\boldsymbol{f}_g^b = \begin{bmatrix} -mg\sin\theta \\ mg\cos\theta\sin\phi \\ mg\cos\theta\cos\phi \end{bmatrix}, \tag{12}$$

and the thrust vector is

$$\boldsymbol{f}_T^b = \begin{bmatrix} 0 & 0 & -F \end{bmatrix}^T. \tag{13}$$

Consequently,

$$\dot{u} = rv - qw - g\sin\theta, \tag{14}$$

$$\dot{v} = pw - ru + g\cos\theta\sin\phi, \tag{15}$$

$$\dot{w} = qu - pv + g\cos\theta\cos\phi - \frac{F}{m}. \tag{16}$$

4.5 Altitude dynamics

Altitude h is positive upward, whereas the inertial down coordinate is $p_d = -h$. From the translational kinematics,

$$\dot{h} = \sin\theta\, u - \sin\phi\cos\theta\, v - \cos\phi\cos\theta\, w. \tag{17}$$

For the simplified inertial model used near hover, differentiation of the position kinematics is performed while the derivative of the rotation matrix and the body-velocity Coriolis terms are neglected. The vertical dynamics then become

$$\ddot{h} = \frac{F}{m}\cos\phi\cos\theta - g. \tag{18}$$

Equation (18) shows that only the vertical component of thrust contributes to altitude acceleration. At level attitude, $\phi = \theta = 0$, and therefore

$$\boxed{\ddot{h} = \frac{F}{m} - g}. \tag{19}$$

Consequently, $F = mg$ corresponds to hover, $F > mg$ produces upward acceleration, and $F < mg$ produces downward acceleration. If the vehicle rolls or pitches while total thrust remains unchanged, the factor $\cos\phi\cos\theta$ decreases. The available vertical thrust is then reduced because part of the thrust vector is redirected into the horizontal plane.

4.6 Rotational dynamics

Let $\boldsymbol{h} = J\boldsymbol{\omega}^b$ denote angular momentum. Euler's rigid-body equation is

$$J\dot{\boldsymbol{\omega}}^b + \boldsymbol{\omega}^b \times (J\boldsymbol{\omega}^b) = \boldsymbol{\tau}^b. \tag{20}$$

Symmetry is used to approximate the inertia tensor by

$$J = \mathrm{diag}(J_x, J_y, J_z). \tag{21}$$

The component equations are

$$\dot{p} = \frac{J_y - J_z}{J_x} qr + \frac{\tau_\phi}{J_x}, \tag{22}$$

$$\dot{q} = \frac{J_z - J_x}{J_y} pr + \frac{\tau_\theta}{J_y}, \tag{23}$$

$$\dot{r} = \frac{J_x - J_y}{J_z} pq + \frac{\tau_\psi}{J_z}. \tag{24}$$

5 Motor and Propeller Model

5.1 Actuator dynamics

Actual rotor speed cannot change instantaneously. A first-order approximation is used for each motor–propeller unit:

$$\dot{\Omega}_i = \frac{\Omega_{i,\mathrm{cmd}} - \Omega_i}{\tau_m}, \qquad \frac{\Omega_i(s)}{\Omega_{i,\mathrm{cmd}}(s)} = \frac{1}{\tau_m s + 1}. \tag{25}$$

This local actuator dynamic does not close the vehicle motion loop: position and attitude are not fed back to compute $\Omega_{i,\mathrm{cmd}}$. The complete vehicle simulation therefore remains open-loop. If experimental evidence shows asymmetric spin-up and spin-down, τ_m may be replaced by τ_{up} or τ_{down} according to the sign of $\Omega_{i,\mathrm{cmd}} - \Omega_i$.

5.2 Thrust and aerodynamic reaction torque

For rotor i, static thrust and reaction-torque magnitudes are approximated as

$$F_i = k_f \Omega_i^2, \qquad Q_i = k_m \Omega_i^2. \tag{26}$$

Thus, doubling rotor speed ideally produces four times the thrust and reaction torque. Beard [**?**] writes the force and torque as linear functions of a calibrated command δ_i. Equation (26) is used here so that the physical quadratic dependence on actual rotor speed is explicit.

6 Control Allocation

The rotor order is front, right, back, and left. Following the propeller rotation directions in Beard [**?**], the mapping from squared rotor speeds to generalized force is

$$\begin{bmatrix} F \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \underbrace{\begin{bmatrix} k_f & k_f & k_f & k_f \\ 0 & -\ell k_f & 0 & \ell k_f \\ \ell k_f & 0 & -\ell k_f & 0 \\ -k_m & k_m & -k_m & k_m \end{bmatrix}}_{\mathcal{M}} \begin{bmatrix} \Omega_f^2 \\ \Omega_r^2 \\ \Omega_b^2 \\ \Omega_l^2 \end{bmatrix}. \tag{27}$$

Equal rotor speeds create collective thrust but cancel all three ideal body moments. A left–right thrust difference produces roll moment, a front–back difference produces pitch moment, and an imbalance between clockwise and counter-clockwise rotor pairs produces yaw moment.

The hover rotor speed follows from $F = mg$:

$$\boxed{\Omega_h = \sqrt{\frac{mg}{4k_f}}}. \tag{28}$$

7 Simulink Implementation

The model is implemented as interconnected nonlinear state-space and actuator subsystems. Four identical motor–propeller subsystems receive commanded rotor speeds. Each subsystem applies (25) and (26). The allocation subsystem implements (27). A MATLAB Function block evaluates the twelve derivatives in (9)–(24), and a twelve-element vector Integrator returns the state $\boldsymbol{x}$ to the nonlinear function.

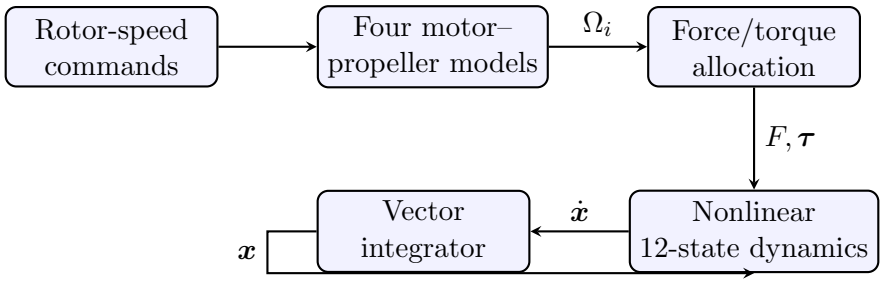


Figure 1: Implemented open-loop Simulink architecture. The state return path performs numerical integration and is not a flight-control feedback loop.

The baseline parameters currently used for model verification are listed in table 1. They represent a generic platform and must not be described as experimentally identified values.

Table 1: Baseline simulation parameters.

Parameter	Value	Unit
m	1.0	kg
g	9.81	$\mathrm{m\,s^{-2}}$
ℓ	0.20	m
J_x, J_y, J_z	$0.01, 0.01, 0.02$	$\mathrm{kg\,m^2}$
k_f	1.0×10^{-5}	$\mathrm{N\,s^2\,rad^{-2}}$
k_m	2.0×10^{-7}	$\mathrm{N\,m\,s^2\,rad^{-2}}$
τ_m	0.08	s
$\Omega_{\max}$	1000	$\mathrm{rad\,s^{-1}}$

For these values, (28) gives $\Omega_h \approx 495.23\,\mathrm{rad\,s^{-1}}$ (approximately 4729 rpm). Each rotor then produces approximately 2.4525 N, giving a total thrust of 9.81 N.

8 Illustration of Reference Frames and State Variables

The geometric meaning of the model is clarified by distinguishing position, orientation, and velocity. Position is measured in an Earth-fixed inertial frame, whereas linear and angular velocities are resolved along the axes attached to the vehicle. The frame sequence below follows Beard [**?**]. The illustrations are reproduced from that reference.

8.1 Inertial and Vehicle Frames

The inertial frame $\mathcal{F}^i$ has its origin at a fixed reference location. Its axes point north, east, and down. The vehicle frame $\mathcal{F}^v$ has its origin at the quadrotor centre of mass, but its axes remain parallel to the inertial axes. Moving from the inertial frame to the vehicle frame therefore changes the origin without changing the orientation. Altitude h is positive upward, so the inertial down coordinate is $-h$.

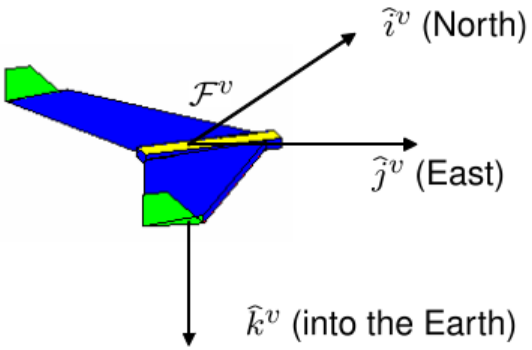


Figure 2: Vehicle frame with its origin at the centre of mass and axes parallel to the inertial axes. Reproduced from Beard [**?**], Figure 5.

8.2 Vehicle Frame to Vehicle-1 Frame: Yaw

The first rotation is a right-hand rotation through yaw angle ψ about the downward axis of $\mathcal{F}^v$. It produces the vehicle-1 frame $\mathcal{F}^{v1}$. The vertical axis is unchanged, while the horizontal axes turn through the heading angle. For zero roll and pitch, the vehicle-1 longitudinal axis points forward along the vehicle.

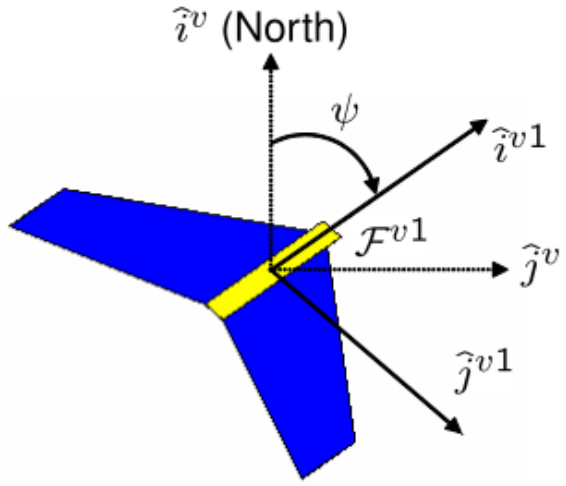


Figure 3: Vehicle-1 frame after the yaw rotation. Reproduced from Beard [**?**], Figure 6.

8.3 Vehicle-1 Frame to Vehicle-2 Frame: Pitch

The second rotation is through pitch angle θ about the lateral axis of $\mathcal{F}^{v1}$. It produces the vehicle-2 frame $\mathcal{F}^{v2}$. The lateral axis is unchanged by this rotation. Positive pitch raises the vehicle nose. When roll is zero, the vehicle-2 frame already coincides with the body frame.

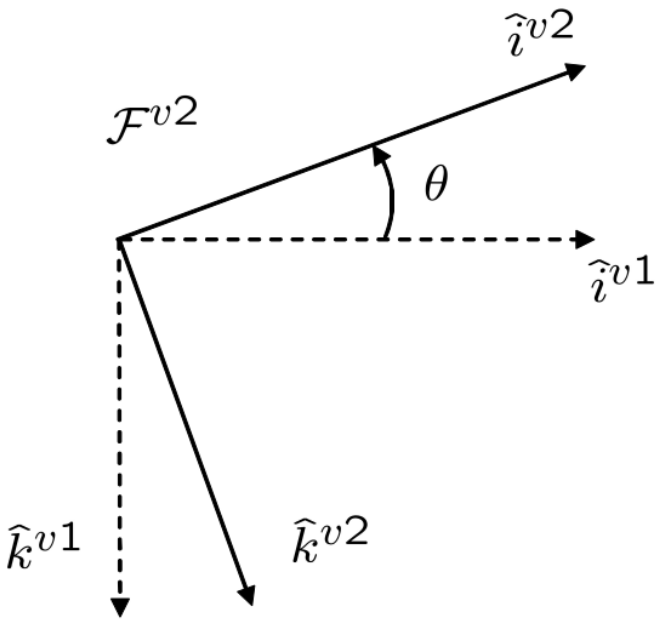


Figure 4: Vehicle-2 frame after the pitch rotation. Reproduced from Beard [**?**], Figure 7.

8.4 Vehicle-2 Frame to Body Frame: Roll

The final rotation is through roll angle ϕ about the longitudinal axis of $\mathcal{F}^{v2}$. It produces the body frame $\mathcal{F}^b$. Its axes point forward (x_b), right (y_b), and downward through the vehicle (z_b). Positive roll lowers the right side. All four moving frames share the same origin at the centre of mass; their orientations differ according to the successive rotations.

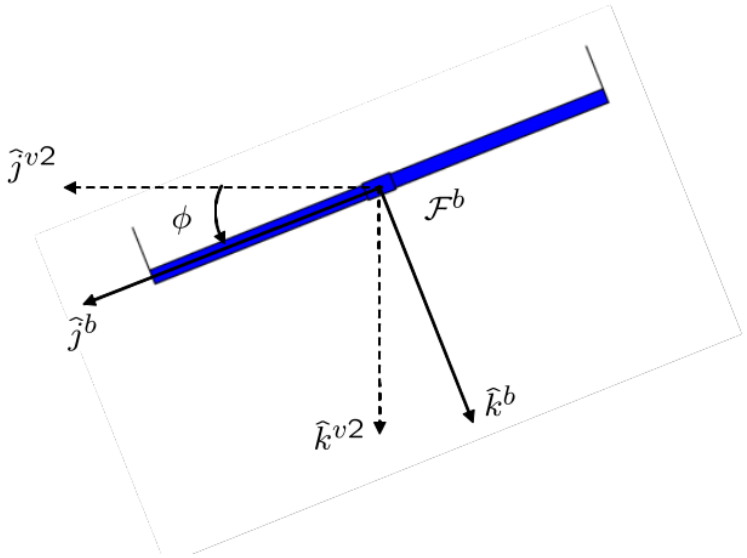


Figure 5: Body frame after the roll rotation. Reproduced from Beard [**?**], Figure 8.

The order is yaw, then pitch, then roll. Each rotation is about an axis of the frame produced by the preceding rotation. Changing this order generally changes the resulting orientation.

8.5 Physical Meaning of the Twelve States

The state vector used in the Simulink model is

$$\boldsymbol{x} = [p_n \; p_e \; h \; u \; v \; w \; \phi \; \theta \; \psi \; p \; q \; r]^T. \tag{29}$$

The individual states are defined as follows:

p_n **(m):** North position relative to the inertial origin.

p_e **(m):** East position relative to the inertial origin.

h **(m):** Altitude above the reference level, positive upward.

u **(m/s):** Linear-velocity component along the forward body axis.

v **(m/s):** Linear-velocity component along the right body axis.

w **(m/s):** Linear-velocity component along the downward body axis.

ϕ **(rad):** Roll angle for the rotation from vehicle-2 to body frame.

θ **(rad):** Pitch angle for the rotation from vehicle-1 to vehicle-2 frame.

ψ **(rad):** Yaw angle for the rotation from vehicle to vehicle-1 frame.

p **(rad/s):** Angular-velocity component about the current forward body axis.

q **(rad/s):** Angular-velocity component about the current right body axis.

r **(rad/s):** Angular-velocity component about the current downward body axis.

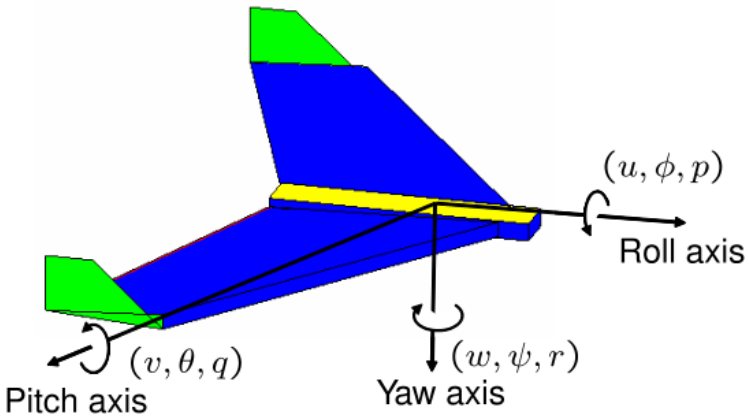


Figure 6: Body axes and associated motion variables. Reproduced from Beard [**?**], Figure 10.

The components u, v, w describe the velocity of the centre of mass resolved in body coordinates; they are not generally the derivatives of p_n, p_e, h. At zero roll, pitch, and yaw, $\dot{p}_n = u$, $\dot{p}_e = v$, and $\dot{h} = -w$.

Likewise, p, q, r describe one angular-velocity vector resolved along the current body axes. The Euler angles parameterize orientation using successive rotations about different axes, so p, q, r are generally not equal to $\dot{\phi}, \dot{\theta}, \dot{\psi}$. Their exact relationship is given by Equation (10).

9 Illustration of Forces and Moments

The baseline model includes gravity, propeller thrust, and propeller reaction torques. Other aerodynamic loads are neglected under the modelling assumptions. Figures 7 and 8 connect these loads to the generalized inputs of the nonlinear model.

9.1 Rotor Arrangement and Rotation Directions

The rotors are ordered front, right, back, and left, each a distance ℓ from the centre of mass. Following the convention shown by Beard, the front and back rotors rotate clockwise when viewed from above, while the right and left rotors rotate counter-clockwise. Each rotor produces an upward thrust and a reaction torque on the body.

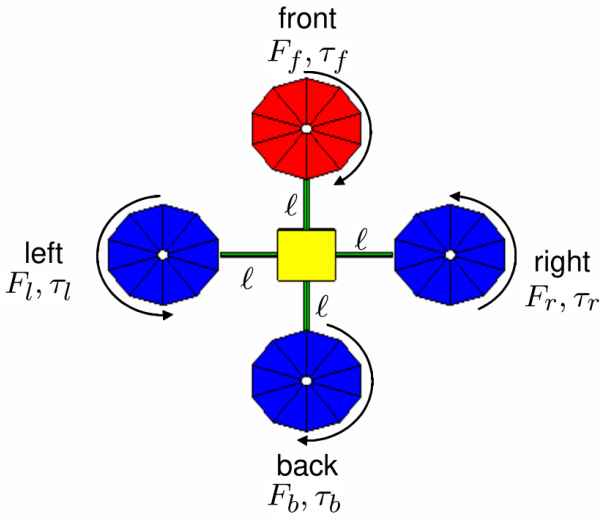


Figure 7: Quadrotor rotor locations and rotation directions. Reproduced from Beard [**?**], Figure 12.

9.2 Collective Thrust and Gravity

The positive thrust magnitude is the sum of the four rotor thrusts:

$$F = F_f + F_r + F_b + F_l, \qquad F_i = k_f \Omega_i^2. \tag{30}$$

The propellers exert force along $-z_b$, so the body-resolved thrust vector is $\boldsymbol{f}_T^b = [0\ 0\ -F]^T$. Gravity acts downward in the inertial frame and has body components

$$\boldsymbol{f}_g^b = \begin{bmatrix} -mg\sin\theta \\ mg\cos\theta\sin\phi \\ mg\cos\theta\cos\phi \end{bmatrix}. \tag{31}$$

For level attitude, $F = mg$ balances gravity. Maintaining a stationary hover also requires zero initial velocity and zero net body moment. Force balance alone does not remove an existing velocity.

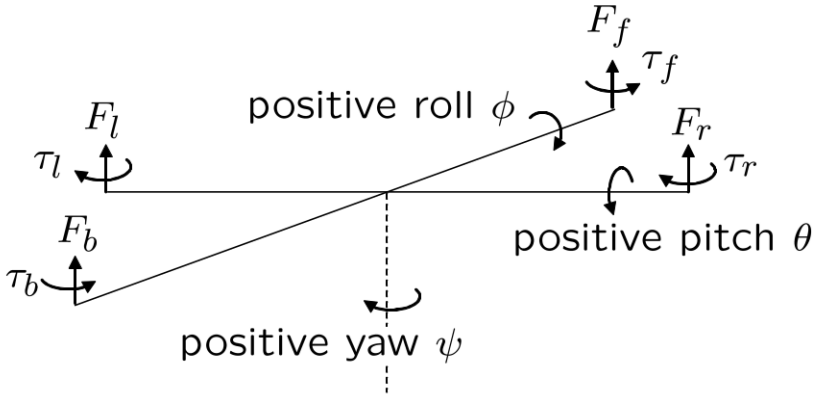


Figure 8: Rotor thrusts, reaction torques, and positive rotation directions. Reproduced from Beard [**?**], Figure 13.

9.3 Roll Moment

The left and right thrusts act on opposite sides of the longitudinal axis. Their difference produces

$$\tau_\phi = \ell(F_l - F_r). \tag{32}$$

A larger left thrust produces a positive roll moment, tending to lower the right side. Equal left and right thrusts cancel this moment.

9.4 Pitch Moment

The front and back thrusts produce a moment about the lateral axis:

$$\tau_\theta = \ell(F_f - F_b). \tag{33}$$

A larger front thrust produces a positive pitch moment, tending to raise the nose. Equal front and back thrusts cancel this moment.

9.5 Yaw Moment

The aerodynamic torque resisting each propeller's rotation produces an opposite reaction torque on the vehicle body. Writing Q_i for the positive reaction-torque magnitude gives

$$Q_i = k_m \Omega_i^2, \qquad \tau_\psi = -Q_f + Q_r - Q_b + Q_l. \tag{34}$$

The Q_i notation corresponds to the individual rotor torque magnitudes labelled τ_i in Beard's illustrations. The signs above follow the rotation directions in Figure 7. Positive yaw is a right-hand rotation about the downward body axis.

Equal speeds of identical rotors cancel the ideal roll, pitch, and yaw moments while retaining collective thrust. Differential speeds generate the body moments represented by the allocation matrix in Equation (27).

10 Open-Loop Simulation Plan and Results

10.1 Thrust versus angular velocity

A sweep of rotor angular velocity verifies the first quadratic static relation in (26). The expected curve begins at the origin and increases parabolically according to $F_i = k_f \Omega_i^2$.

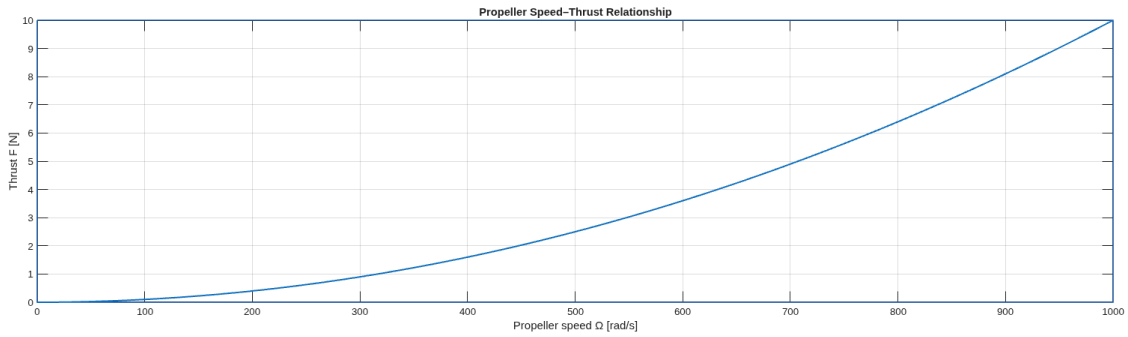


Figure 9: Propeller thrust as a function of actual propeller angular velocity.

Figure 10: Enter Caption

10.2 Reaction torque versus angular velocity

The same angular-velocity sweep verifies $Q_i = k_m \Omega_i^2$. The plotted quantity is the positive torque magnitude; its signed contribution to vehicle yaw depends on whether the propeller rotates clockwise or counter-clockwise.

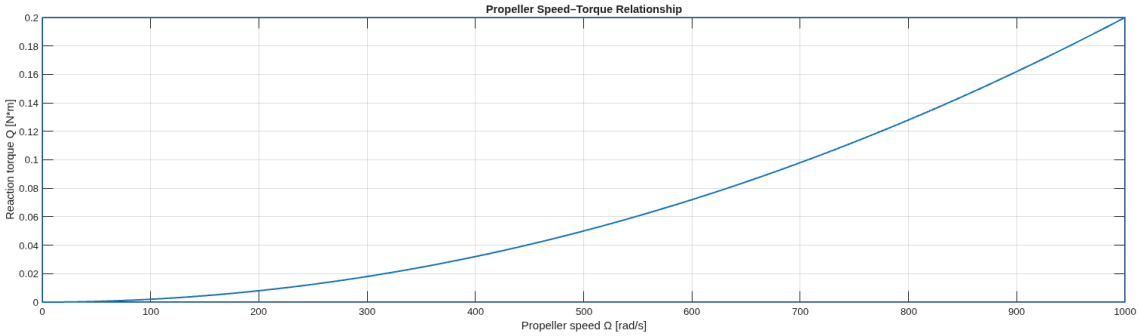


Figure 11: Propeller aerodynamic reaction-torque magnitude as a function of actual propeller angular velocity. Its signed contribution to vehicle yaw depends on the rotor rotation direction.

10.3 Motor transient

A commanded-speed step verifies the actuator time constant and illustrates that thrust rises according to the square of the actual, delayed rotor speed.

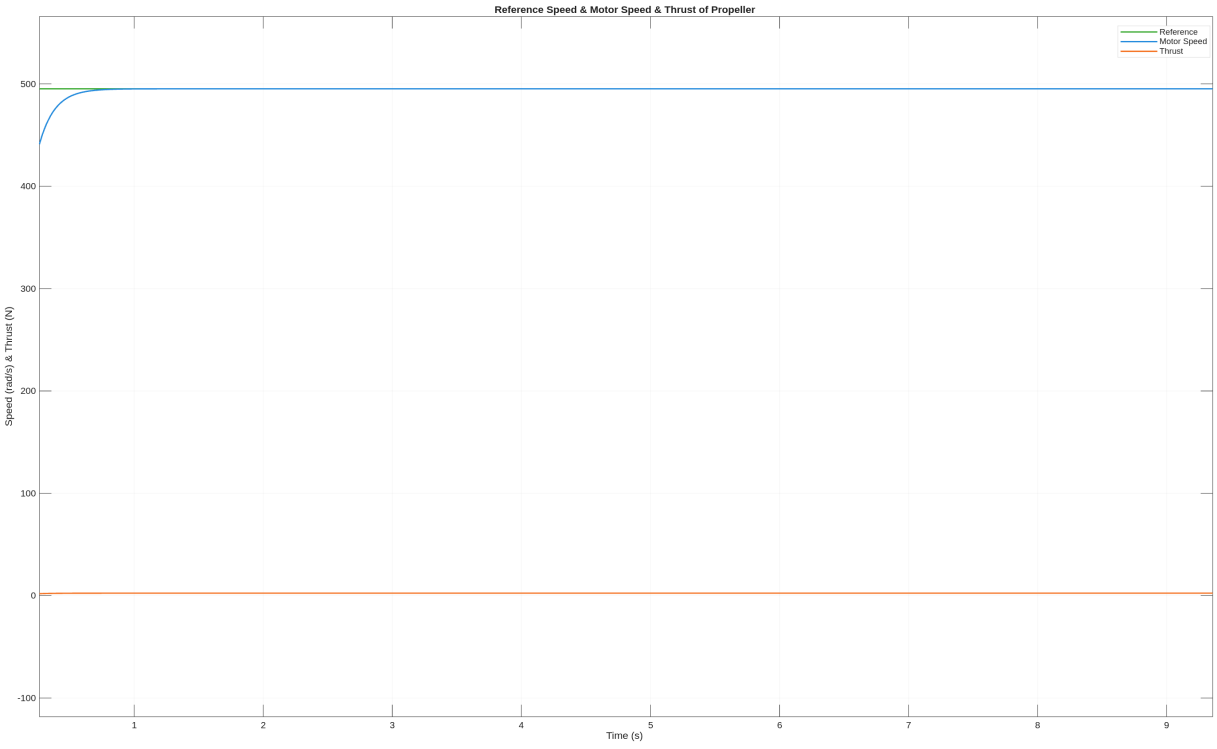


Figure 12: Commanded rotor speed, actual rotor speed, and resulting propeller thrust during the motor transient test.

10.4 Hover equilibrium and collective input

With all four rotors at Ω_h, the expected generalized input is

$$F = mg, \qquad \tau_\phi = \tau_\theta = \tau_\psi = 0. \tag{35}$$

At level attitude this produces zero acceleration. If all speeds are multiplied by a common factor α, then $F = \alpha^2 mg$ and the initial upward acceleration is

$$a_{\text{up}} = (\alpha^2 - 1)g. \tag{36}$$

For example, $\alpha = 1.05$ gives approximately $1.01\,\text{m}\,\text{s}^{-2}$. In an ideal open-loop model, a constant thrust greater than weight produces increasing vertical velocity rather than a constant climb rate.

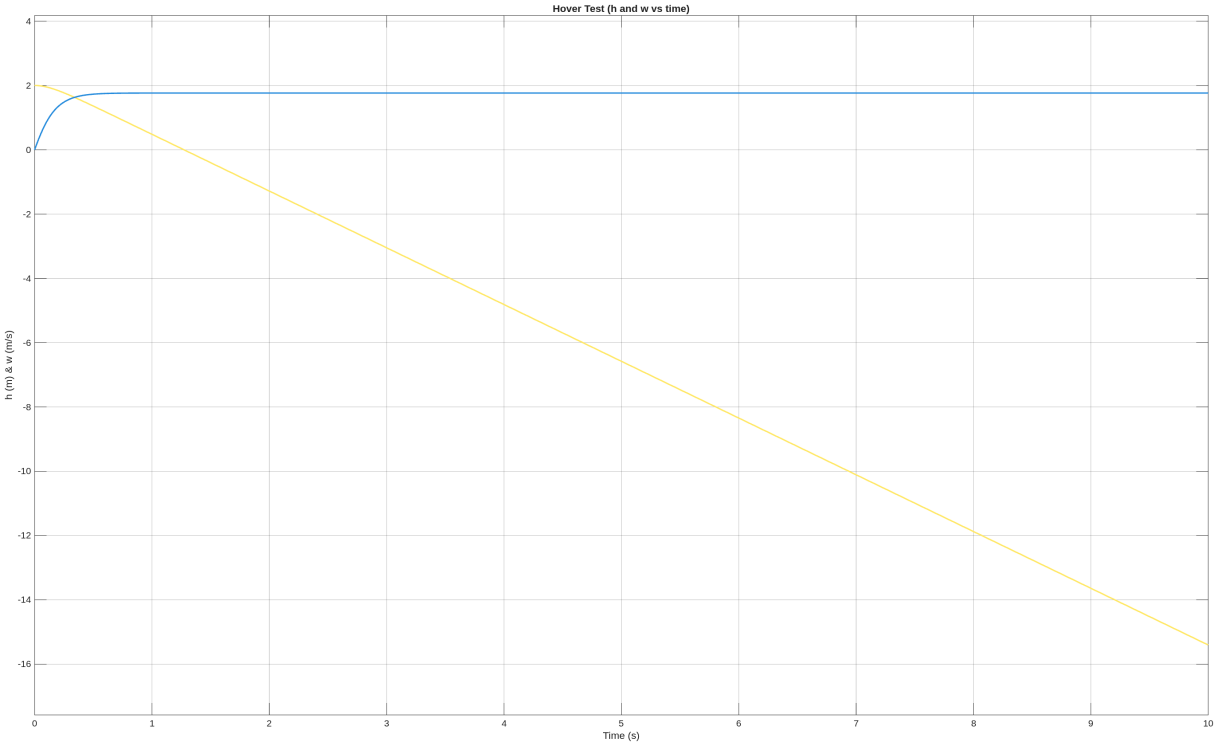


Figure 13: Open-loop altitude and body vertical velocity under nominal hover-speed commands.

10.5 Pitch Input and Longitudinal Motion

This test investigates how a front–back motor-speed difference generates pitch moment and affects vehicle attitude, horizontal position, and altitude. The motor commands are applied directly, without attitude or position feedback. The existing first-order motor dynamics remain active.

10.5.1 Motor Commands

The front motor reference is increased above the hover speed, while the back motor reference is decreased by the same amount. The left and right references remain at the hover value:

$$\Omega_{f,\text{cmd}} = \Omega_h + \Delta\Omega, \tag{37}$$
$$\Omega_{b,\text{cmd}} = \Omega_h - \Delta\Omega, \tag{38}$$
$$\Omega_{r,\text{cmd}} = \Omega_h, \tag{39}$$
$$\Omega_{l,\text{cmd}} = \Omega_h, \tag{40}$$

where $\Delta\Omega = 1\,\mathrm{rad\,s^{-1}}$ and $\Omega_h \approx 495.23\,\mathrm{rad\,s^{-1}}$. The simulation duration is 3 s.

The short duration limits the attitude excursion: a sustained open-loop moment produces continued rotation rather than convergence to a commanded pitch angle.

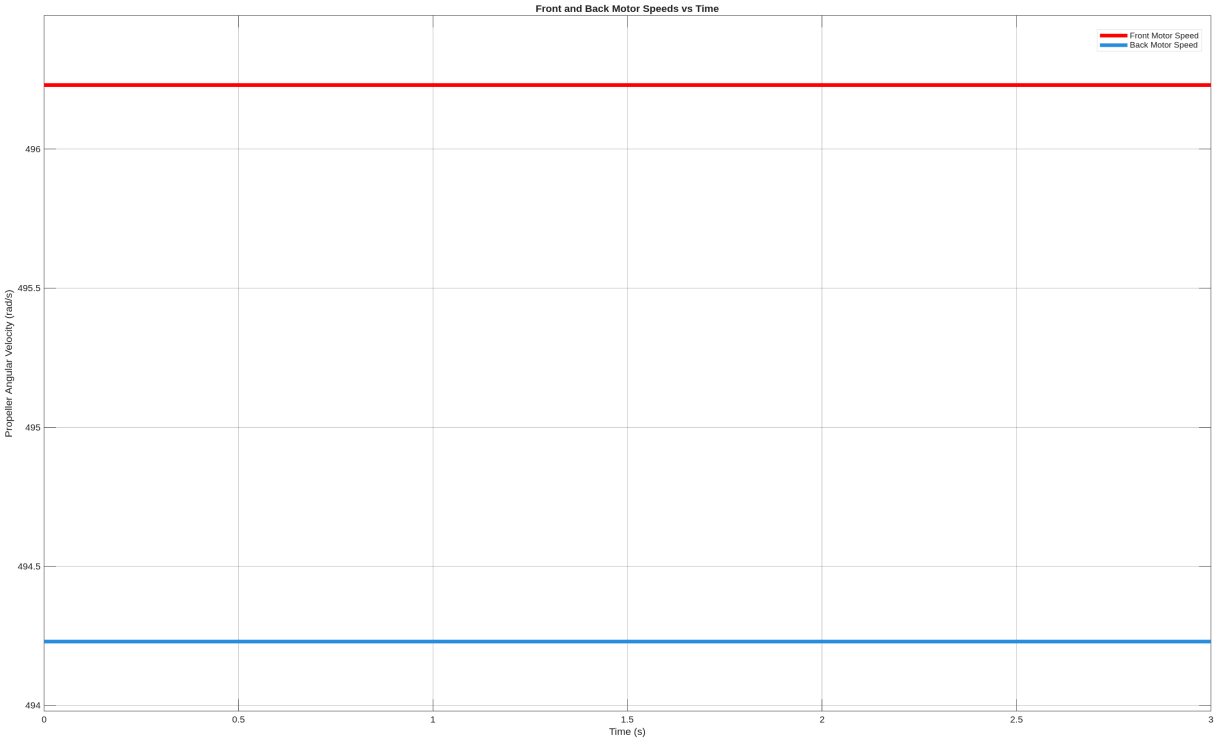


Figure 14: Actual front and back motor speeds during the pitch-input test, in radians per second.

10.5.2 Pitch Moment

The pitch moment depends on the actual rotor speeds:

$$\tau_\theta = \ell(F_f - F_b) = \ell k_f(\Omega_f^2 - \Omega_b^2). \tag{41}$$

The greater front thrust produces a positive pitch moment, tending to raise the nose. Once the motor speeds approach their references, the expected moment is

$$\tau_{\theta,\mathrm{ss}} = 4\ell k_f \Omega_h \Delta\Omega \approx 3.962 \times 10^{-3}\,\mathrm{N\,m}. \tag{42}$$

The collective thrust remains approximately equal to the weight. Its steady-state value is $F_{\mathrm{ss}} = mg + 2k_f(\Delta\Omega)^2$. The small residual yaw moment is $-2k_m(\Delta\Omega)^2$, so this configuration predominantly excites pitch.

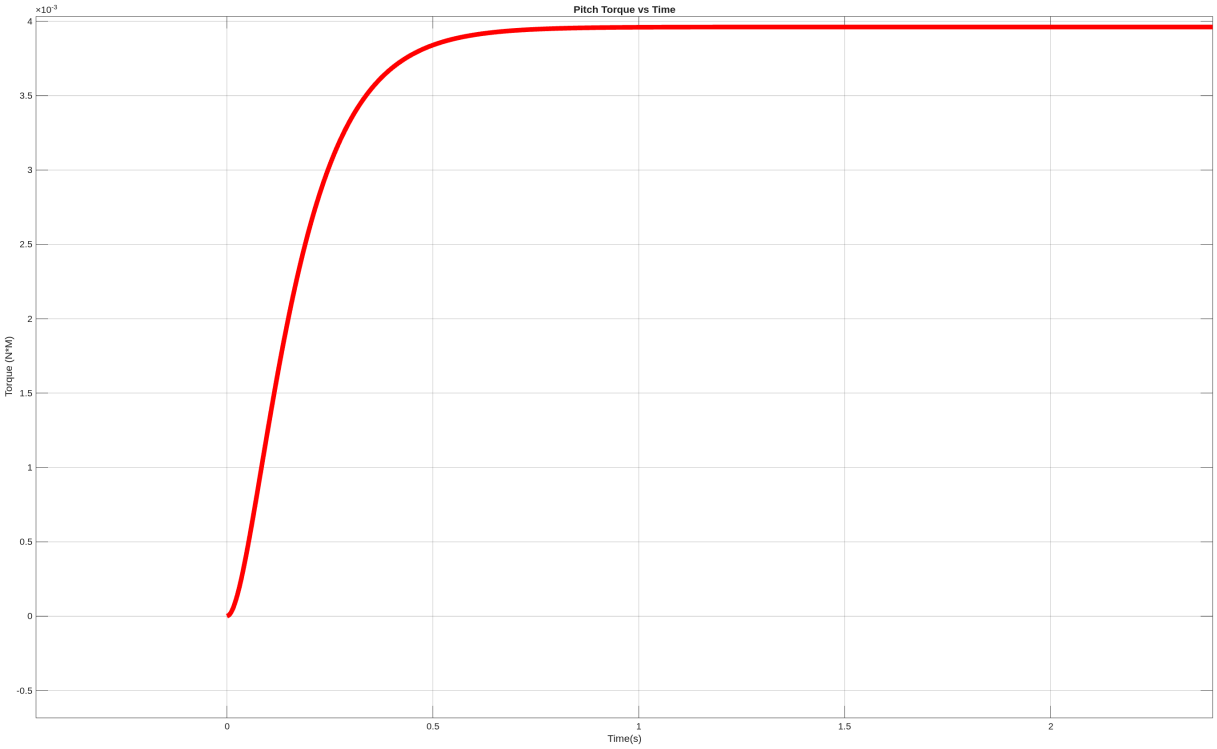


Figure 15: Pitch moment generated by the front–back thrust difference, in newton metres.

10.5.3 Pitch Rate and Pitch Angle

The body pitch-rate dynamics are

$$\dot{q} = \frac{J_z - J_x}{J_y} pr + \frac{\tau_\theta}{J_y}. \tag{43}$$

When roll and yaw rates remain small, the applied pitch moment dominates. A sustained positive moment therefore produces an approximately linear increase in q after the actuator transient.

The pitch-angle kinematics are

$$\dot{\theta} = q \cos\phi - r \sin\phi. \tag{44}$$

For predominantly pitch motion with small roll, $\dot{\theta} \approx q$. The pitch angle is consequently expected to grow approximately quadratically after the initial transient. The angle θ is measured in radians, whereas q is measured in radians per second.

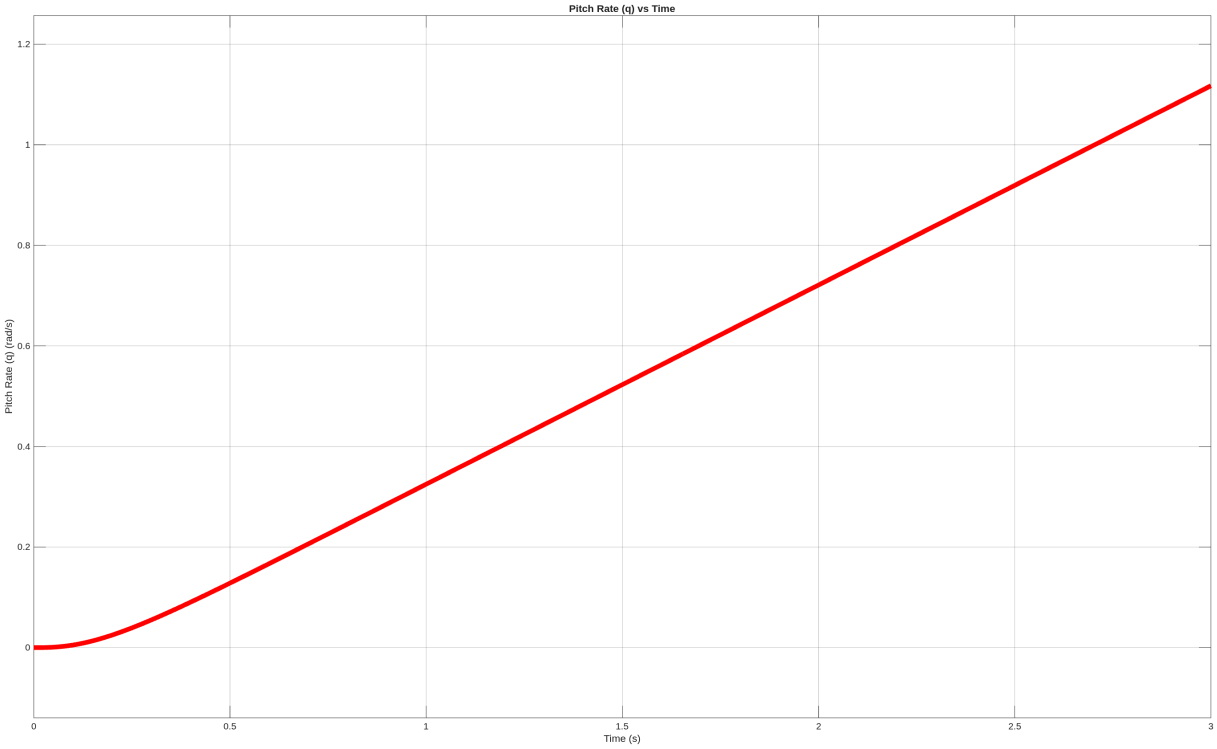


Figure 16: Body pitch rate q during the pitch-input test, in radians per second.

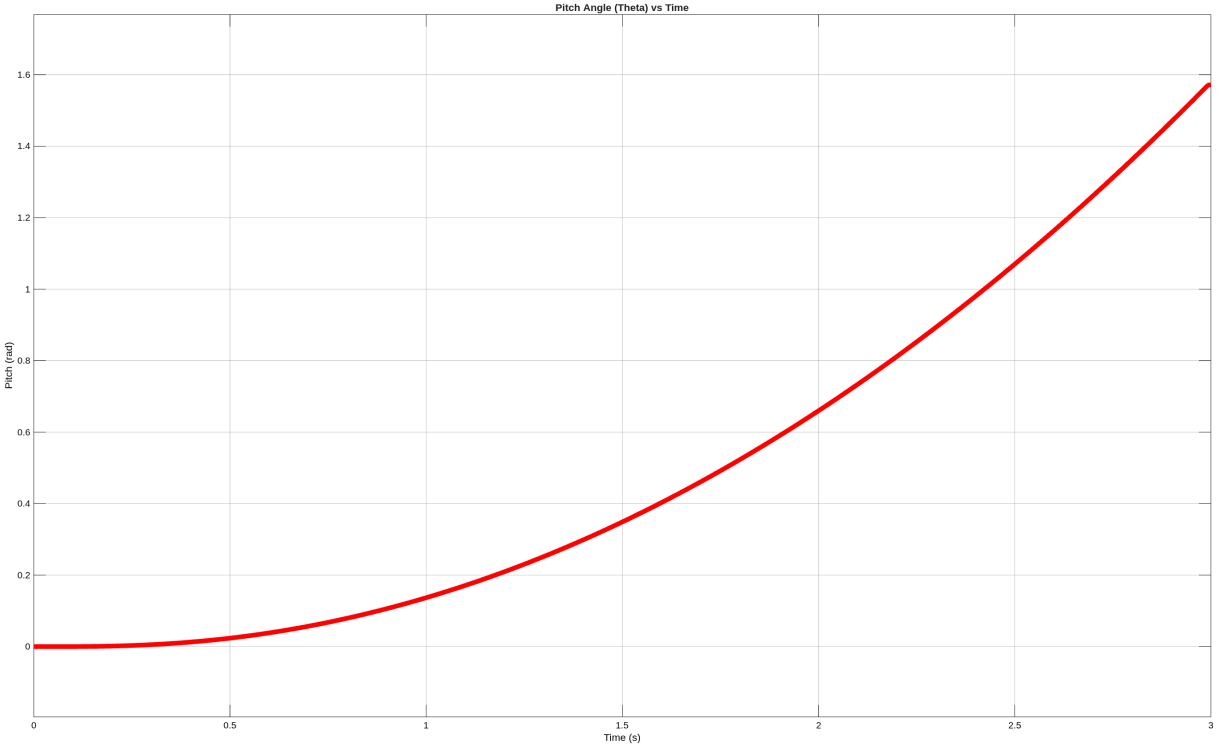


Figure 17: Pitch angle θ during the pitch-input test, in radians.

10.5.4 Horizontal Motion and Altitude

Pitch changes the direction of the thrust vector. For negligible roll and yaw, the inertial accelerations reduce to

$$\ddot{p}_n = -\frac{F}{m}\sin\theta, \qquad \ddot{h} = \frac{F}{m}\cos\theta - g. \tag{45}$$

Under the adopted coordinate convention, positive pitch produces acceleration toward negative north position p_n. As the pitch angle grows, the vertical component of thrust decreases. Since total thrust remains close to the weight, this reduction is expected to cause altitude loss.

Altitude h is positive upward, so decreasing h indicates descent. If the motors start from zero speed, the initial thrust deficit also contributes to descent. That startup effect must be distinguished from altitude loss caused by tilting.

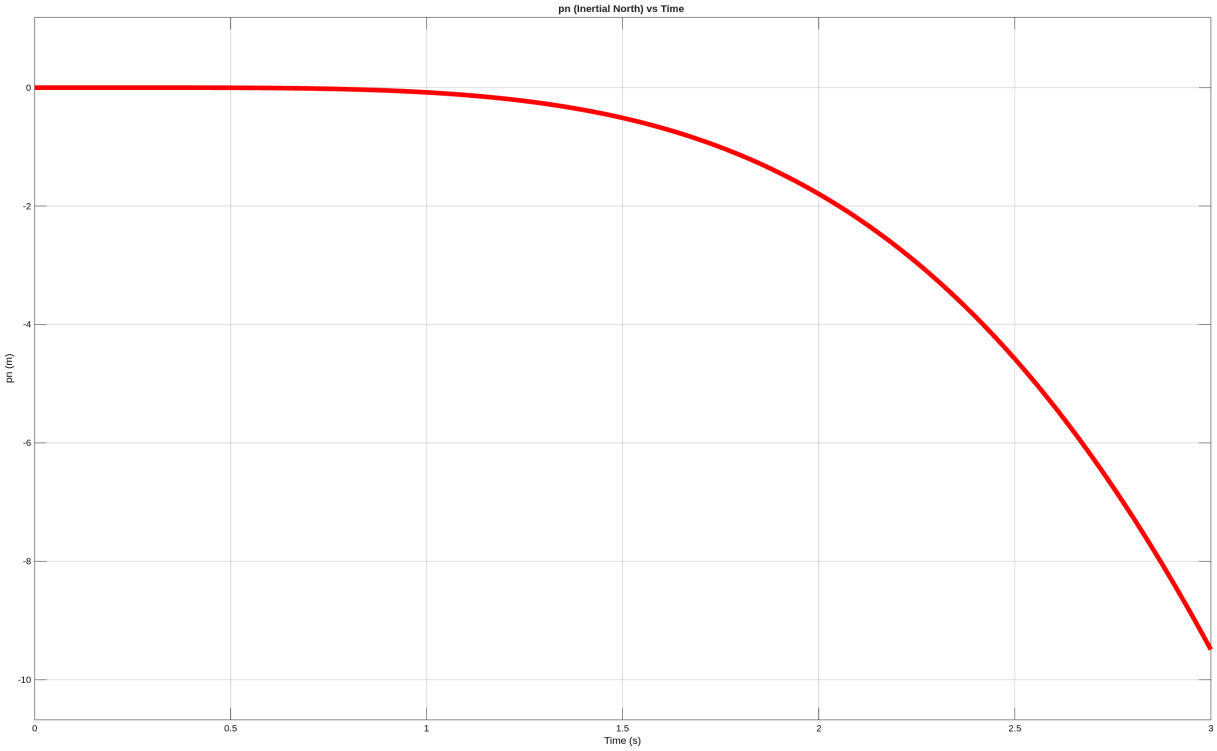


Figure 18: North position p_n during the pitch-input test, in metres.

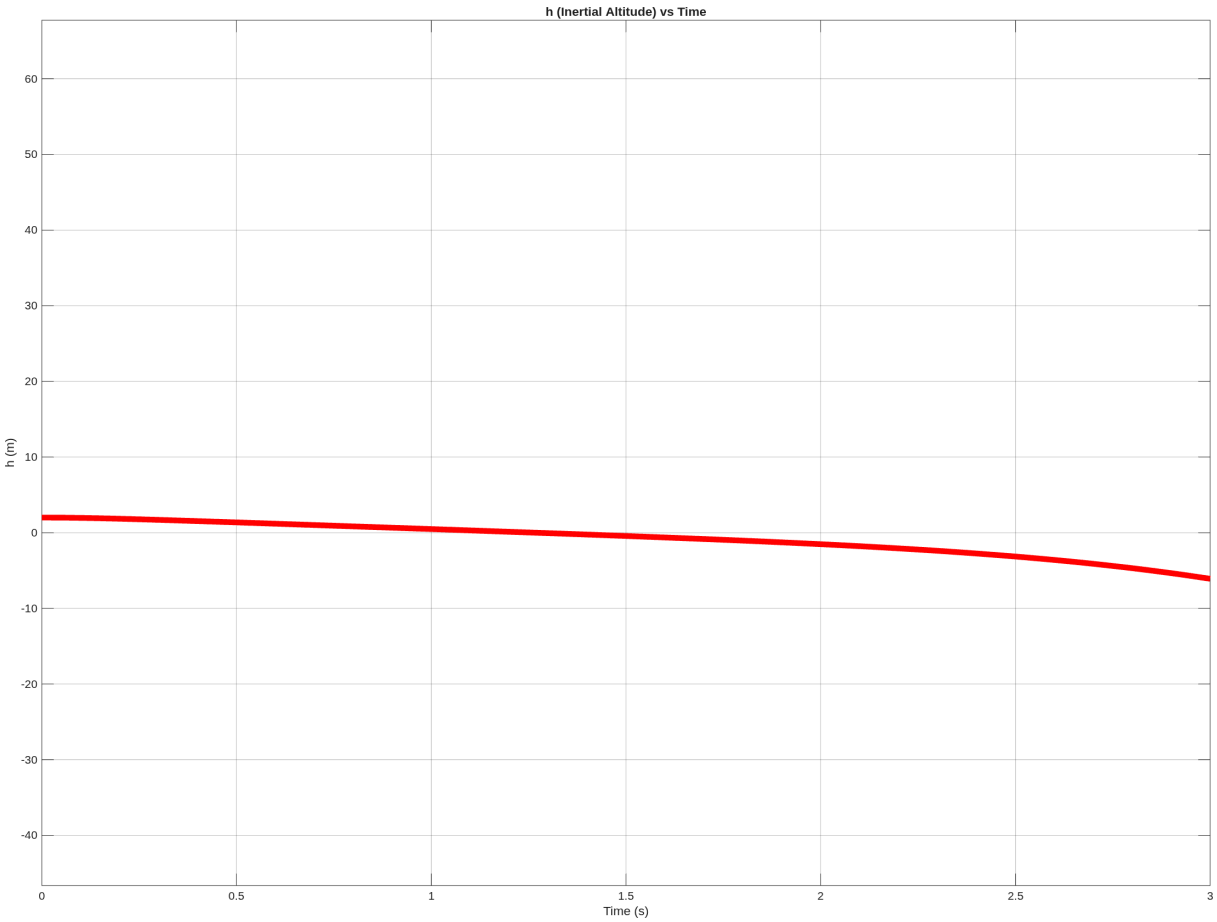


Figure 19: Altitude h during the pitch-input test, in metres, with positive direction upward.

11 MATLAB/Simulink and Gazebo Co-Simulation

The nonlinear quadrotor model implemented in MATLAB/Simulink was compared with a separate Gazebo simulation. The same four motor-speed references were supplied to both implementations. Simulink integrated the analytical equations of motion, while Gazebo simulated the vehicle using its physics engine.

ROS 2 provided the communication interface between the two environments. Motor commands were transmitted from Simulink to Gazebo, and Gazebo odometry was returned to Simulink for state extraction, logging, and comparison.

The purpose of this experiment was to examine consistency between the analytical model and the Gazebo implementation. The results represent a comparison between two simulations; experimental validation against a physical vehicle remains a separate task.

11.1 Software Environment and Gazebo Setup

The implementation used Ubuntu 24.04, ROS 2 Jazzy, Gazebo Sim 8.11.0, and MATLAB/Simulink R2026a.

The project was organized inside a ROS workspace containing the package `gazebo_matlab_cosim`. This package contained the custom motor-command adapter and the Gazebo simulation assets.

The Gazebo implementation included a world file and a quadrotor model description. The world file, `empty_world.sdf`, included the quadrotor at an initial altitude of 2 m. The model description defined the vehicle links, joints, inertial properties, rotor systems, and odometry

publisher.

The environment variable `GZ_SIM_RESOURCE_PATH` was configured to include the model directory. This allowed Gazebo to locate the quadrotor referenced by the world file.

Before launching the ROS-related processes, the ROS installation and the compiled project workspace were sourced. The participating ROS 2 processes used domain ID zero. The domain ID selects the ROS discovery domain, while topic names and message types identify the individual data streams.

Separate terminals were used to launch MATLAB, Gazebo, the ROS–Gazebo bridge, and the motor-command adapter. The saved models and compiled workspace were retained between sessions. Therefore, unchanged project files did not require rebuilding whenever the simulation was reopened.

11.2 Communication Architecture

The communication system consisted of a command path and a measurement path.

The command path transmitted motor-speed references from Simulink to Gazebo through the following components:

1. Simulink ROS 2 Publish block;
2. custom motor-command adapter;
3. ROS–Gazebo parameter bridge; and
4. Gazebo motor systems.

The measurement path returned Gazebo odometry through the bridge to the Simulink ROS 2 Subscribe block.

11.2.1 Motor-Command Topic

Simulink published the four motor-speed references on

`/quadrotor/matlab_motor_commands`

using the message type

`std_msgs/msg/Float64MultiArray`.

The message contained four numerical values, corresponding to the four motor-speed references.

11.2.2 Motor-Command Adapter

The custom adapter was launched using the executable `motor_command_adapter`.

It subscribed to the MATLAB command topic and copied the four received references into the `velocity` array of an `Actuators` message.

The converted message was published on

`/quadrotor/command/motor_speed`

using the message type

`actuator_msgs/msg/Actuators`.

The adapter performed message-format conversion. It did not generate a feedback-control action or integrate the vehicle equations of motion.

11.2.3 ROS–Gazebo Bridge

The `ros_gz_bridge` parameter bridge translated between ROS 2 messages and Gazebo Transport messages.

For motor commands, the configured message pair was `actuator_msgs/msg/Actuators` and `gz.msgs.Actuators`.

For odometry, the configured pair was `nav_msgs/msg/Odometry` and `gz.msgs.Odometry`.

The launch command enabled bidirectional bridging. The directions used in this experiment were motor commands towards Gazebo and odometry towards Simulink.

11.3 Simulink Command Blocks

11.3.1 Motor References and Mux

The four motor-speed references were combined using a Mux block:

$$\Omega_{\text{ref}} = \begin{bmatrix} \Omega_{1,\text{ref}} \\ \Omega_{2,\text{ref}} \\ \Omega_{3,\text{ref}} \\ \Omega_{4,\text{ref}} \end{bmatrix}. \tag{46}$$

The reference signals branched before the analytical model's motor dynamics. One branch drove the local motor models, while the other supplied the Gazebo command path.

Therefore, the transmitted values were motor-speed references, rather than the actual speeds calculated by the local actuator model. Each simulation branch represented its own actuator response.

11.3.2 Zero-Order Hold

A Zero-Order Hold block sampled the transmitted command vector with a sample period of

$$T_s = 0.05 \text{ s}. \tag{47}$$

The corresponding nominal command frequency was

$$f_s = \frac{1}{T_s} = 20 \text{ Hz}. \tag{48}$$

The sampled reference was held constant between updates, providing a discrete command signal for the ROS interface.

11.3.3 Blank Message and Bus Assignment

A ROS 2 Blank Message block supplied the `Float64MultiArray` message bus.

A Bus Assignment block inserted the four-element command vector into the message's `data` field. The valid array length was specified by assigning `uint32(4)` to

`data_SL_Info.CurrentLength`.

The configured array capacity was four elements.

Compatible discrete sample times were assigned to the message source, command vector, and array-length signal before they entered the nonvirtual message bus.

11.3.4 ROS 2 Publish

The ROS 2 Publish block transmitted the assembled message to the motor-command adapter. The adapter converted its format, and the bridge forwarded the resulting command to Gazebo.

11.4 Odometry Reception and State Extraction

Gazebo odometry was returned on

`/quadrotor/odometry`

using the message type `nav_msgs/msg/Odometry`.

A ROS 2 Subscribe block received this topic with a Simulink sample period of 0.05 s.

Bus Selector blocks extracted the position, orientation quaternion, linear velocity, angular velocity, and message timestamp.

A MATLAB Function block converted these measurements into the state representation used by the analytical model. Workspace logging blocks recorded the following signals inside the simulation output object:

- `simu_states`: analytical model states;
- `gaz_states`: converted Gazebo states;
- `gz_time`: Gazebo message timestamps.

11.5 Coordinate Conventions

The analytical state vector was

$$x = [p_n,\ p_e,\ h,\ u,\ v,\ w,\ \phi,\ \theta,\ \psi,\ p,\ q,\ r]^{\mathrm{T}}. \tag{49}$$

Altitude h is positive upward. The body frame has forward, right, and downward axes. The variables u, v, and w are the linear-velocity components along these body axes.

The variables ϕ, θ, and ψ are the roll, pitch, and yaw Euler angles using the ZYX convention. The variables p, q, and r are the angular-velocity components about the body axes.

The Euler-angle derivatives are related to the body angular rates through the attitude kinematic equations. The Euler angles are therefore not generally obtained by independently integrating p, q, and r.

The project associated Gazebo world coordinates with the analytical position variables through

$$p_n = x_{\mathrm{G}}, \qquad p_e = -y_{\mathrm{G}}, \qquad h = z_{\mathrm{G}}. \tag{50}$$

The orientation quaternion was converted into a rotation matrix. Frame transformations were applied before extracting the Euler angles.

For the level-attitude vertical test, the vertical velocity conversion reduces to

$$w = -v_{z,\mathrm{G}}, \qquad \dot{h} = -w. \tag{51}$$

Thus, positive w indicates descent, while negative w indicates ascent in this test.

Vertical motion alone cannot verify every coordinate transformation. When the vehicle rotates, the distinction between world-frame and body-frame velocity must also be checked.

11.6 Execution and Timestamp Alignment

Gazebo was initially kept paused while the bridge, adapter, and Simulink model were prepared. Simulink was started, and Gazebo was then unpaused.

The Simulink stop time was 10 s, with simulation pacing enabled at a nominal factor of one. The simulators did not use lockstep synchronization. Their available recorded simulation intervals were therefore not assumed to be identical.

11.6.1 Message Timestamps

The Gazebo odometry timestamp was reconstructed from its seconds and nanoseconds fields:

$$t_{\mathrm{G}} = t_{\mathrm{sec}} + 10^{-9} t_{\mathrm{nanosec}}. \tag{52}$$

This timestamp represents the time associated with the Gazebo measurement. It differs from the Simulink time at which the subscriber output was logged.

Repeated subscriber outputs can contain the same timestamp when no new message has arrived. Consequently, the number of logged Simulink samples does not necessarily equal the number of distinct Gazebo measurements.

11.6.2 Comparison Procedure

The recorded data were processed as follows:

1. Uninitialized outputs with zero Gazebo timestamps were discarded.
2. Backward timestamp jumps were checked to detect resets.
3. One observation was retained for each distinct Gazebo timestamp.
4. The comparison was restricted to the common available simulation-time interval.
5. Analytical Simulink states were linearly interpolated at the retained Gazebo timestamps.
6. State errors were calculated using the aligned observations.

No fitted time shift, time-axis scaling, extrapolation, or trajectory smoothing was applied.

11.6.3 Error Measures

For state component x_j, the error was defined as

$$e_j(t_k) = x_{j,\mathrm{G}}(t_k) - x_{j,\mathrm{S}}(t_k), \tag{53}$$

where the Simulink result was interpolated at the Gazebo timestamp t_k.

The root-mean-square error was

$$\mathrm{RMSE}_j = \sqrt{\frac{1}{N}\sum_{k=1}^{N} e_j(t_k)^2}. \tag{54}$$

The maximum absolute error was

$$E_{j,\max} = \max_{1\le k\le N} \left| e_j(t_k) \right|. \tag{55}$$

Each retained timestamp contributed one equally weighted observation.

11.7 Equal-Motor Vertical-Motion Test

All four motor-speed references were set to

$$\Omega_{i,\mathrm{ref}} = 520 \text{ rad/s}, \qquad i = 1, 2, 3, 4. \tag{56}$$

The initial altitude was 2 m. All other initial vehicle states and the initial actual motor speeds were zero.

Equal references were selected to examine vertical motion without intentionally applying differential roll, pitch, or yaw moments.

11.7.1 Expected Response

For the analytical model,

$$m = 1 \text{ kg}, \qquad g = 9.81 \text{ m/s}^2, \tag{57}$$

and the thrust coefficient was

$$k_f = 10^{-5} \text{ N/(rad/s)}^2. \tag{58}$$

The nominal hover speed was therefore

$$\Omega_h = \sqrt{\frac{mg}{4k_f}} \approx 495.23 \text{ rad/s}. \tag{59}$$

The applied reference of 520 rad/s was above the nominal hover speed. After the actuator transient, the analytical model's total thrust approached

$$F = 4k_f(520)^2 = 10.816 \text{ N}. \tag{60}$$

This exceeds the vehicle weight of 9.81 N.

Because the motors initially started from rest, the vehicle first descended while thrust developed. The descent was subsequently arrested, and the vehicle began to climb.

This experiment was an open-loop vertical-motion test rather than a stationary hover-equilibrium test.

11.7.2 Recorded Timing and Startup Anomaly

The repeated run ended at a Simulink time of 10.000 s and a final logged Gazebo timestamp of 8.710 s.

There were 176 distinct positive Gazebo timestamps and no backward timestamp jumps.

The shorter recorded Gazebo interval does not, by itself, establish a constant difference between the simulator execution rates. The recorded interval can also be affected by the delay before Gazebo is unpaused and before messages are received.

Extreme vertical-velocity values appeared in the earliest odometry samples. At a Gazebo timestamp of 0.010 s, the converted vertical velocity was approximately

$$w_{\mathrm{G}} = -222.16 \text{ m/s}, \tag{61}$$

while the altitude remained approximately 1.9995 m.

At the next retained timestamp, 0.060 s, the vertical-velocity difference was approximately 0.02768 m/s.

The isolated early values were treated as a startup odometry anomaly. Their precise software cause was not established.

The raw observations were preserved. The reported comparison was explicitly restricted to the interval

$$0.060 \text{ s} \leq t_{\mathrm{G}} \leq 8.710 \text{ s}. \tag{62}$$

11.7.3 Comparison Results

The altitude and body vertical-velocity responses are presented separately in

The comparisons use Gazebo message timestamps. Analytical Simulink states are interpolated at these timestamps when calculating the errors. Observations before 0.060 s are excluded from the reported comparison, while the raw data are preserved.

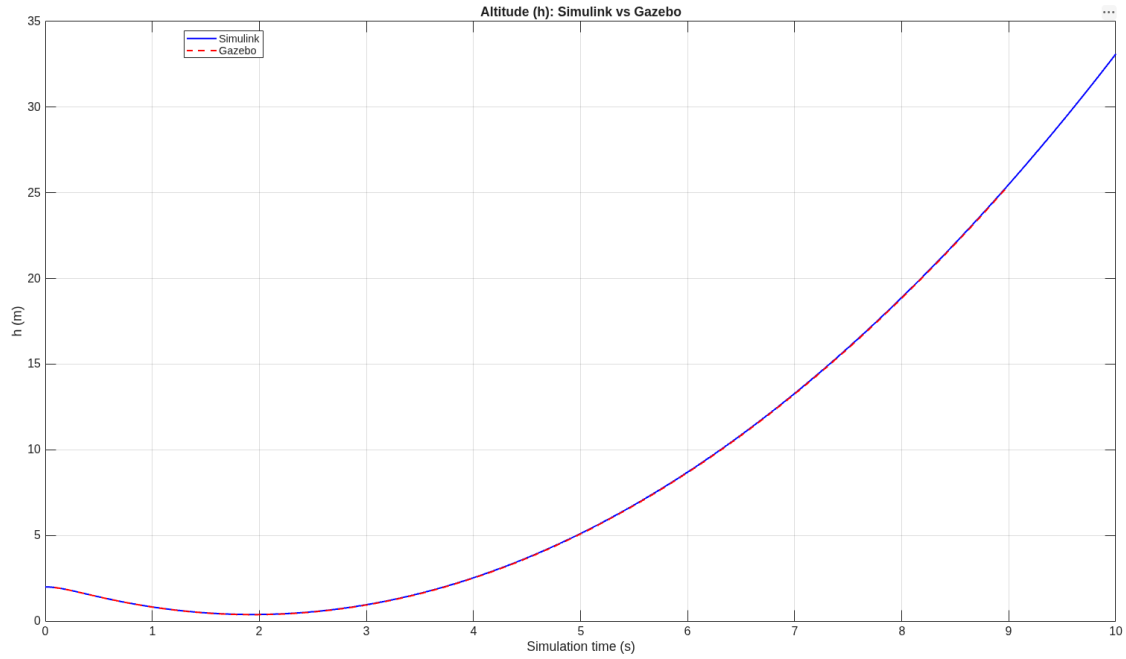


Figure 20: Altitude comparison between Simulink and Gazebo for four equal motor-speed references of 520 rad/s. Altitude h is positive upward.

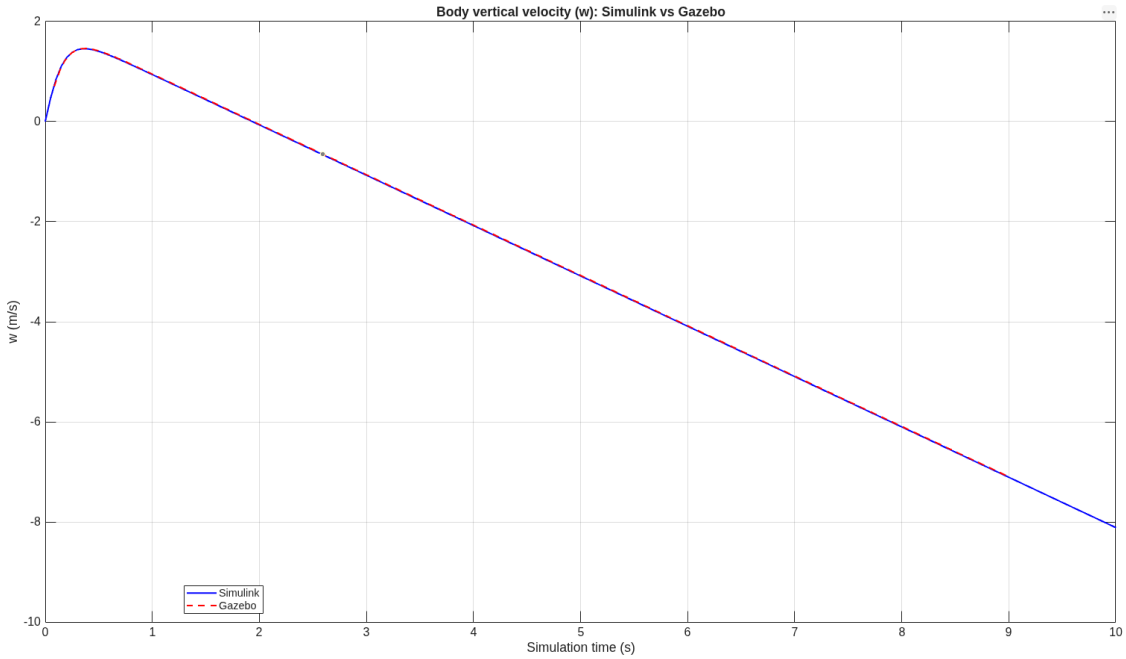


Figure 21: Body vertical-velocity comparison between Simulink and Gazebo for four equal motor-speed references of 520 rad/s. Positive w indicates downward motion at level attitude. Gazebo observations before 0.060 s are excluded.

The numerical comparison is summarized in Table 2. The error is defined as the Gazebo state minus the interpolated Simulink state. Both metrics are evaluated over the same retained

time interval.

Table 2: Vertical-state errors over the common comparison interval, excluding Gazebo timestamps below 0.060 s.

State	RMSE	Maximum absolute error
Altitude h (m)	0.025823	0.044337
Body vertical velocity w (m/s)	0.009887	0.019257

The altitude RMSE was approximately 2.58 cm, with a maximum absolute error of 4.43 cm. The body vertical-velocity RMSE was approximately 0.00989 m/s, with a maximum absolute error of 0.01926 m/s.

These results indicate close agreement between the analytical Simulink model and Gazebo for the tested equal-motor vertical response over the retained comparison interval.

11.8 Pitch-Input Co-Simulation Test

The second co-simulation test examined the response to a front–back motor-speed difference. The objective was to compare pitch rotation and its coupling to longitudinal and vertical motion in Simulink and Gazebo.

The quantities compared were pitch angle θ, body pitch rate q, north position p_n, and altitude h.

11.8.1 Test Configuration

The command-array order was verified from the Gazebo motor plugins as front, right, back, and left. These correspond to Gazebo motor numbers 0, 1, 2, and 3.

The same motor-speed references were supplied to both simulation branches:

$$\Omega_{\text{ref}} = \begin{bmatrix} \Omega_{f,\text{ref}} \\ \Omega_{r,\text{ref}} \\ \Omega_{b,\text{ref}} \\ \Omega_{l,\text{ref}} \end{bmatrix} = \begin{bmatrix} 520.2 \\ 520.0 \\ 519.8 \\ 520.0 \end{bmatrix} \text{ rad/s.} \tag{63}$$

The baseline reference was $\Omega_0 = 520\,\text{rad/s}$, and the differential reference was $\Delta\Omega = 0.2\,\text{rad/s}$. The references were applied from the start of the simulation.

Initial altitude was 20 m in both models. The other initial vehicle states and the initial actual motor speeds were zero.

Table 3: Pitch-input co-simulation configuration.

Setting	Value
Baseline reference Ω_0 (rad/s)	520
Differential reference $\Delta\Omega$ (rad/s)	0.2
Initial altitude (m)	20
Initial actual motor speeds (rad/s)	0
Other initial vehicle states	0
Simulink stop time (s)	5
Command and subscriber sample period (s)	0.05
Motor time-constant setting (s)	0.12
Retained comparison interval (s)	0.005 to 4.125

The Simulink stop time specifies the local integration duration. The Gazebo observations cover the interval actually received and logged during the run. These intervals need not have identical end times because the simulators were not synchronized in lockstep.

11.8.2 Consistent Whole-Vehicle Inertia

An initial comparison showed a difference in pitch response because the same inertia values had been assigned to the complete analytical vehicle and to the Gazebo base link alone.

In Gazebo, the four rotor links add their own inertias and contributions due to their positions. The analytical model must account for these contributions when representing the same assembled vehicle.

The Gazebo base-link mass remained 0.96 kg, and each rotor-link mass was 0.01 kg. The total vehicle mass was therefore 1 kg.

The original Gazebo base-link inertias were retained:

$$J_{x,\text{base}} = J_{y,\text{base}} = 0.01, \qquad J_{z,\text{base}} = 0.02 \quad (\text{kg m}^2). \tag{64}$$

For the simplified single-body analytical representation, the rotor contributions were included using the parallel-axis theorem. The calculated whole-vehicle inertias about the total centre of mass were

$$J_x = J_y = 0.01085536, \qquad J_z = 0.02168 \quad (\text{kg m}^2). \tag{65}$$

These values were used in the analytical model for the pitch-input comparison.

The adjustment was calculated from the masses, inertias, and rotor positions in the Gazebo model description. It was not obtained by fitting the parameters to the observed trajectories.

Gazebo retains articulated rotor links, whereas the analytical plant uses an equivalent rigid-body representation.

11.8.3 Expected Response

The front–back thrust difference generates the analytical pitch moment

$$\tau_\theta = \ell k_f \left(\Omega_f^2 - \Omega_b^2\right). \tag{66}$$

At the commanded steady speeds,

$$\tau_\theta = 4\ell k_f \Omega_0 \Delta\Omega = 8.32 \times 10^{-4} \text{ N m}. \tag{67}$$

For small roll and yaw rates,

$$\dot{q} \approx \frac{\tau_\theta}{J_y} \approx 0.0766 \text{ rad/s}^2. \tag{68}$$

The pitch-angle kinematics satisfy

$$\dot{\theta} = q\cos\phi - r\sin\phi. \tag{69}$$

For predominantly pitch motion, $\dot{\theta} \approx q$. A sustained positive pitch moment therefore produces an approximately linear increase in pitch rate and an approximately quadratic increase in pitch angle after the motor transient.

The symmetric motor-speed offsets introduce a small second-order change in collective thrust:

$$F = 4k_f\Omega_0^2 + 2k_f(\Delta\Omega)^2 \approx 10.816 \text{ N}. \tag{70}$$

The front and back rotors share the same rotation direction in the Gazebo model. Their squared-speed changes therefore also produce a small second-order reaction-torque imbalance. The input predominantly excites pitch and is not an exactly pure pitch moment.

For negligible roll and yaw, the analytical inertial accelerations are

$$\ddot{p}_n = -\frac{F}{m}\sin\theta, \qquad \ddot{h} = \frac{F}{m}\cos\theta - g. \tag{71}$$

Positive pitch therefore produces motion towards negative north under the adopted convention. Tilting reduces the vertical component of thrust.

The initial descent also reflects the motor startup transient, since the actual motor speeds initially start from zero.

11.8.4 Data Processing

The comparison used the Gazebo odometry timestamps. Outputs with zero timestamps were discarded, backward timestamp jumps were checked, and one observation was retained per distinct valid timestamp.

The analytical states were linearly interpolated at the retained Gazebo timestamps within the common available simulation-time interval.

For this pitch comparison, no additional fixed startup-time cutoff was applied after removing zero-timestamp outputs.

No fitted time shift, time-axis scaling, extrapolation, or trajectory smoothing was used.

11.8.5 Pitch Angle Comparison

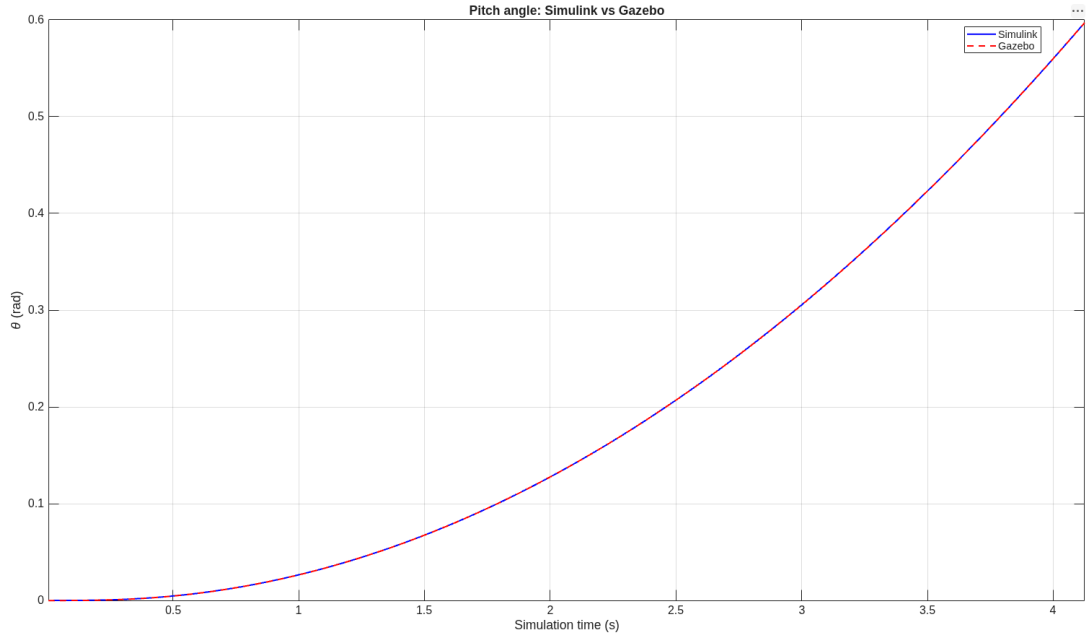


Figure 22: Simulink and Gazebo pitch angle θ for the front–back differential motor-speed input. Angles are in radians.

The pitch-angle curves closely overlap over the displayed interval. Both show an increasing positive pitch angle, consistent with the applied positive pitch moment.

The recorded pitch response remains below $\pi/2$, avoiding the ZYX pitch singularity and the branch change associated with inverse-sine pitch extraction.

11.8.6 Pitch Rate Comparison

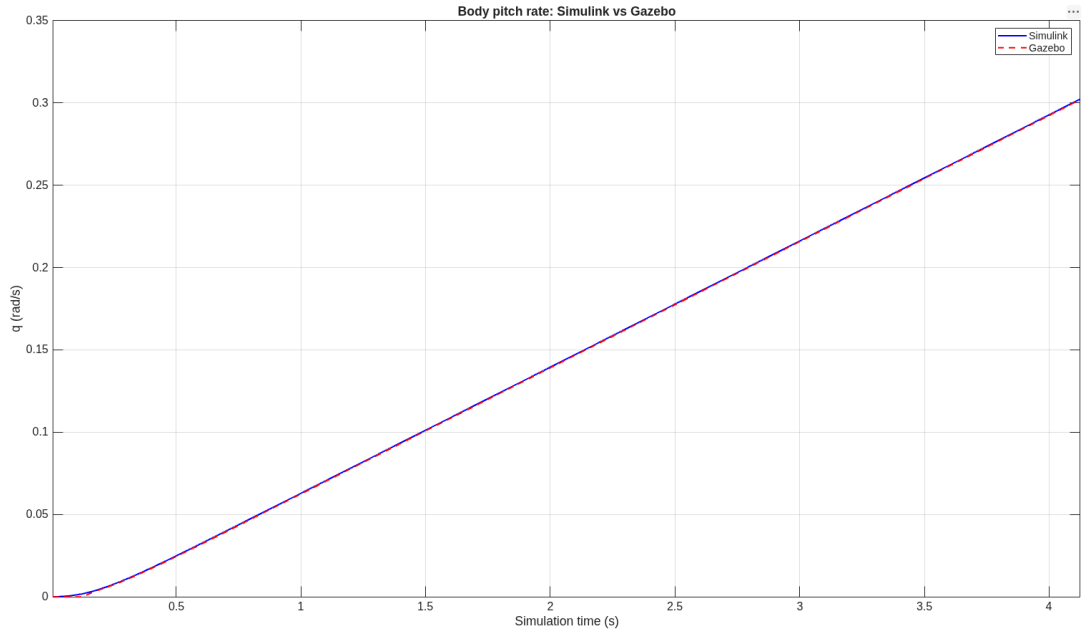


Figure 23: Simulink and Gazebo body pitch rate q, expressed in radians per second, for the pitch-input test.

The pitch-rate responses also closely overlap. Following the initial motor transient, both increase approximately linearly.

Together with the pitch-angle comparison, this supports consistency of the tested pitch response after accounting for the rotor-link inertia contributions.

11.8.7 Longitudinal Position Comparison

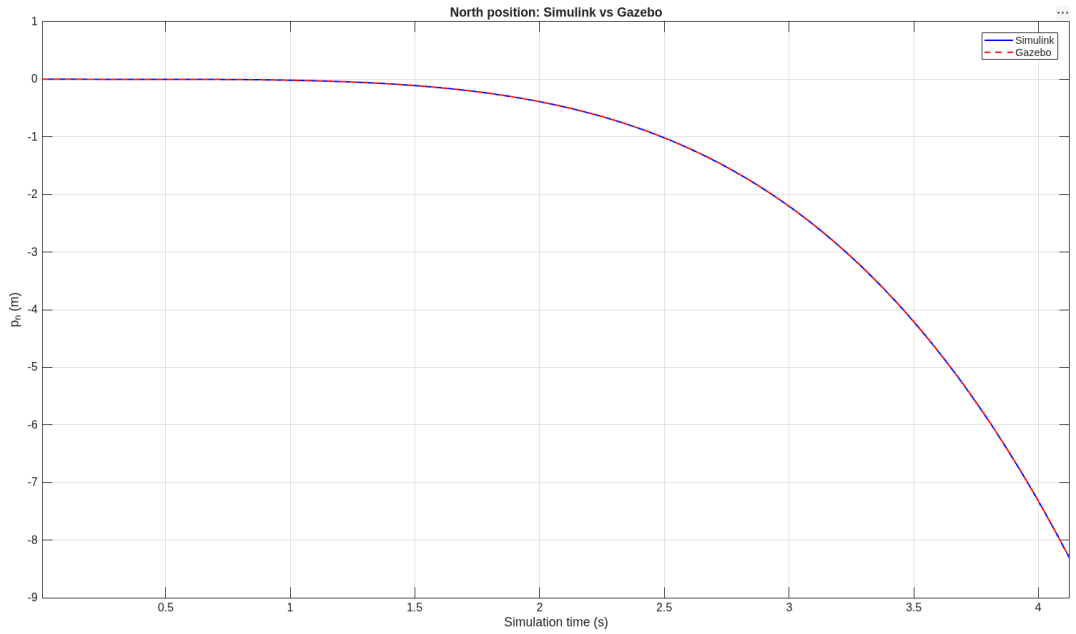


Figure 24: Simulink and Gazebo north position p_n during the pitch-input test. Position is expressed in metres.

Both implementations predict displacement towards negative north. The magnitude of the displacement grows as the increasing pitch angle redirects a larger component of thrust horizontally.

The two position curves closely overlap over the displayed interval.

11.8.8 Altitude Comparison

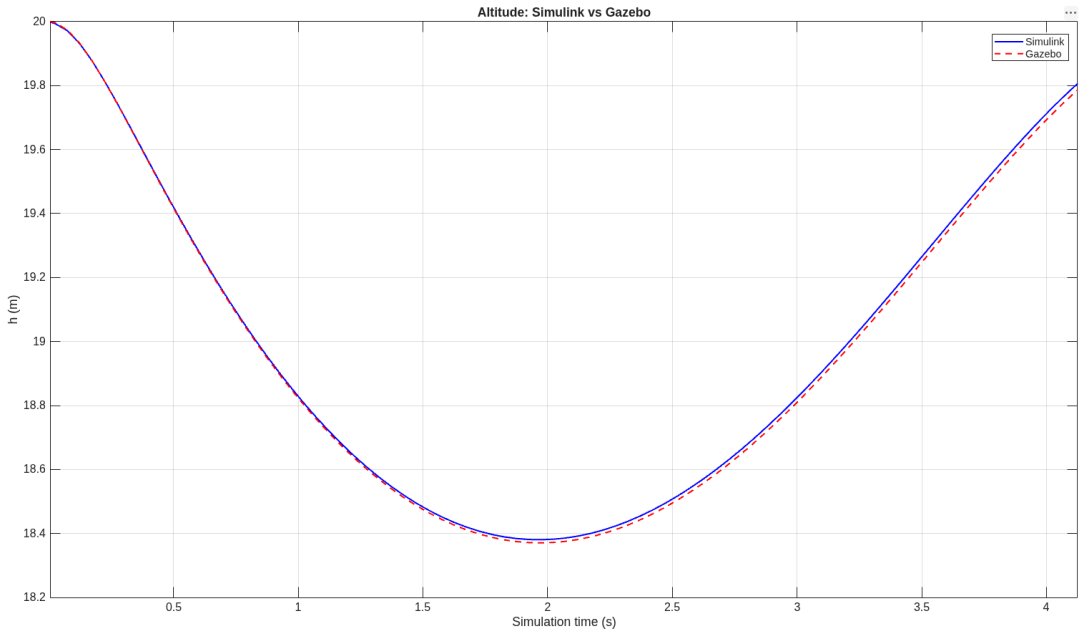


Figure 25: Simulink and Gazebo altitude h during the pitch-input test, starting at 20 m. Altitude is positive upward.

Both models predict an initial altitude reduction while motor thrust builds, followed by recovery during the displayed interval.

A small altitude difference remains visible, with the Gazebo result slightly below the analytical result towards the end of the plotted interval.

The vehicle remains above the ground, so floor contact does not enter the displayed comparison.

11.8.9 Numerical Comparison

Errors are defined as Gazebo states minus the interpolated analytical states. RMSE and maximum absolute error are evaluated using the same retained timestamps for all four quantities.

Table 4: Pitch-input co-simulation errors over the retained comparison interval.

State	RMSE	Maximum absolute error
Pitch angle θ (rad)	2.3057×10^{-5}	2.3953×10^{-5}
Body pitch rate q (rad/s)	4.2427×10^{-4}	1.7937×10^{-3}
North position p_n (m)	0.0010628	0.0023509
Altitude h (m)	0.011749	0.019825

The pitch-angle RMSE was 2.3057×10^{-5} rad, and the pitch-rate RMSE was 4.2427×10^{-4} rad/s.

The north-position RMSE was approximately 1.06 mm, with a maximum absolute difference of approximately 2.35 mm.

The altitude RMSE was approximately 1.17 cm, with a maximum absolute difference of approximately 1.98 cm.

Together, the graphical and numerical comparisons show close agreement in the tested pitch rotation and its coupling to longitudinal and vertical motion after using consistent whole-vehicle inertia values.